

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

(Start the day with the name of Allah)

Welcome To Virtual
University



VU KMS

اسلام علیکم:

امید کرتا ہوں سب خیریت سے ہوں گئے۔ آنے والے تمام نیو طالب علم کو میں ورچوئل یونیورسٹی آف پاکستان میں خوش آمدید کرتا ہوں۔
اگر کسی سٹوڈنٹ کو کسی بھی قسم کی ہیلپ چاہے ہو تو ہم سے رابطہ کر سکتا ہے جیسا کہ: پاسٹ پیپرز، کورس سلیکشن، پروجیکٹ سلیکشن، ڈیٹ شیٹ،
فیس پے اور چینج سٹڈی پروگرام وغیرہ۔

Paid

VU LMS

اگر کوئی سٹوڈنٹ کسی مجبوری کی وجہ سے یا جاب ہو لڈر ہونے کی وجہ سے اپنا ایل ایم ایس خود ہینڈل نہیں کر سکتا تو وہ ہم سے پیڈ سروس حاصل کر
سکتا ہے کوشش کرے کے آپ ہر چیز خود اٹھیمپٹ کریں اگر نہیں کر سکتے تو سروس حاصل کریں۔

Email:

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انشاء اللہ آپ ہماری سروس سے مطمئن ہوں گے۔

Whatsapp: 03066295623 , 03404643855

مزید اگر کسی رہنمائی کی ضرورت ہو تو وہ رابطہ کر سکتا ہے۔

Question # 9 of 30 (Start time: 12:15:13 PM, 22 June 2021)

If a function has recursive call as the last statement, it is known as _____

Select the correct option

☐	Local Recursion
☐	Last recursion
☒	Tail recursion
☐	Function recursion

Solved By: Waqas Ahmed

Whatsapp: 03066295623

Email: waqas.vu.edu@gmail.com

Question # 10 of 30 (Start time: 12:15:28 PM, 22 June 2021)

Analyze the given code carefully and identify which type of binary tree traversal this is :

```
void traversal (TreeNode<int>* treeNode)
{
    if( treeNode != NULL )
    {
        traversal (treeNode->getLeft());
```

Select the correct option

☐	Preorder
☐	Postorder
☒	Inorder
☐	Levelorder

Solved By: Waqas Ahmed
Whatsapp: 03066295623
Email: waqas.vu.edu@gmail.com



Question # 11 of 30 (Start time: 12:15:44 PM, 22 June 2021)

If we use singly linked list to implement list, then there is an issue that it gives difficulty when we:

Select the correct option



Move backward in the list

Solved By: Waqas Ahmed

Whatsapp: 03066295623

Email: waqas.vu.edu@gmail.com



We will decrease its size



Move forward in the list



We will increase its size

Question # 12 of 30 (Start time: 12:16:04 PM, 22 June 2021)

The _____ of a node in a binary tree is defined as the height of its left subtree minus height of its right subtree.

Select the correct option



Height



Balance

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com



Level



Width

Question # 13 of 30 (Start time: 12:16:21 PM, 22 June 2021)

What is the postfix form of $(A+B)*(C-D)$

Select the correct option



$CD - A B + *$



$A + B * C - D$



$A B + C D - *$

Solved By: Waqas Ahmed

Whatsapp: 03066295623

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$+A B - C D *$

Question # 14 of 30 (Start time: 12:16:38 PM, 22 June 2021)

Suppose a Stack class has been defined using template. Now, we want to declare a Stack object of an int type. What will be the correct syntax?

Select the correct option



int Stack stack ;



<int>Stack stack ;



Stack int stack;



Stack<int> stack;

Solved By: Waqas Ahmed

Whatsapp: 03066295623

Email: waqas.vu.edu@gmail.com

Question # 15 of 30 (Start time: 12:16:51 PM, 22 June 2021)

If both left and right nodes of a node are NULL then this type of node is called a _____ node.

Select the correct option



Non leaf



Leaf

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com



Inner



Internal

Question # 16 of 30 (Start time: 12:17:08 PM, 22 June 2021)

The first statement in the given code is called as:

```
class BinarySearchTree ;  
class BinaryNode{  
friend class BinarySearchTree;  
};
```

Select the correct option



Post Class Declaration



Pre Class Declaration



Forward Declaration

Solved By: Waqas Ahmed

Whatsapp: 03066295623

Email: waqas.vu.edu@gmail.com



Binary Declaration

Question # 17 of 30 (Start time: 12:17:24 PM, 22 June 2021)

Head node pointer always points to the _____ of the linked list?

Select the correct option



current node



newly added node



first node

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com



last node



Question # 18 of 30 (Start time: 12:17:38 PM, 22 June 2021)

_____ parameter passing (by value or by reference) is similar to PASCAL.

Select the correct option



FORTRAN



C++

Solved By: Waqas Ahmed

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COBOL



JAVA

Question # 19 of 30 (Start time: 12:17:57 PM, 22 June 2021)

In singly linked list a node consists of two parts:

Select the correct option



Object and pointer

Solved By: Waqas Ahmed

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Object and structure



Two pointers



Two objects

Question # 20 of 30 (Start time: 12:18:13 PM, 22 June 2021)

In Preorder traversal of a binary tree the second step is _____

Select the correct option

☐	traverse the right subtree
☐	traverse the leaf node
☐	traverse the root
☒	traverse the left subtree

Solved By: Waqas Ahmed
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Email: waqas.vu.edu@gmail.com

Question # 21 of 30 (Start time: 12:18:27 PM, 22 June 2021)

At a particular node, the difference in heights of its left and right subtree gives the _____ of the node

Select the correct option



Right subtree



Height



Left subtree



Balance

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com



Question # 22 of 30 (Start time: 12:18:43 PM, 22 June 2021)

Binary Search Tree violates the condition of AVL tree when any node has balance factor equal to

Select the correct option

☐	-1	
☒	2 or -2	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	0	
☐	1	

Question # 23 of 30 (Start time: 12:19:41 PM, 22 June 2021)

In a complete binary tree the number of nodes at level 5 are

Select the correct option

☐	15	
☐	31	
☐	16	
☒	32	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com

Question # 24 of 30 (Start time: 12:20:21 PM, 22 June 2021)

Local variables defined inside function body are _____ automatically at the end of function execution.

Select the correct option

☒	destroyed	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	created	
☐	incremented	
☐	decremented	

Question # 25 of 30 (Start time: 12:20:39 PM, 22 June 2021)

Each node in doubly link list has,

Select the correct option

☐	4 pointers	
☒	2 pointers	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	1 pointer	
☐	3 pointers	



Question # 26 of 30 (Start time: 12:20:58 PM, 22 June 2021)

Analyze the given code carefully and identify which type of binary tree traversal this is :

```
void traversal (TreeNode<int>* treeNode)
{
    if( treeNode != NULL )
    {
        cout << *(treeNode->getInfo())<<" ";
    }
}
```

Select the correct option



Inorder



Preorder

Solved By: Waqas Ahmed
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Sorted order



Postorder

Question # 27 of 30 (Start time: 12:21:24 PM, 22 June 2021)

In a strictly complete binary tree, the number of _____ at any level k will be 2^k .

Select the correct option



links



nodes

Solved By: Waqas Ahmed

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sets



children

Question # 28 of 30 (Start time: 12:21:48 PM, 22 June 2021)

_____ is the maximum height of the AVL tree.

Select the correct option

 $\log_2 n$ 

1.44

 $4.44 \log_2 n$  $1.44 \log_2 n$

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com

Question # 29 of 30 (Start time: 12:22:04 PM, 22 June 2021)

A binary tree is said to be a _____ binary tree if every non-leaf node in a binary tree has non-empty left and right subtrees.

Select the correct option

☐	Complete
☒	Strictly
☐	AVL
☐	Perfect

Solved By: Waqas Ahmed
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Question # 30 of 30 (Start time: 12:22:25 PM, 22 June 2021)

Consider the linked list having data [6, 72, 35, 65, 25] stored in it. While current pointer is pointing to memory location having 35 stored in it. What will be the resultant linked list after calling add(18) function on the given linked list.

Solved By: Waqas Ahmed

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Select the correct option



6, 72, 35, 65, 25, 18



6, 72, 18, 35, 65, 25



6, 72, 35, 18, 65, 25



6, 18, 72, 35, 65, 25

Question # 1 of 30 (Start time: 12:10:58 PM, 22 June 2021)

Total Marks: 1

After deletion of a node from a binary search tree , _____ traversal method should be maintained.

Select the correct option

[Reload Math Equations](#)

Level order



Inorder

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com



Preorder



Postorder

Question # 2 of 30 (Start time: 12:11:34 PM, 22 June 2021)

Total Marks: 1

If there are _____ nodes in an AVL tree, its levels will be roughly as $\text{Log}_2(10 \text{ million})$.

Select the correct option



5 million



100 million



2 million



10 million

Solved By: Waqas Ahmed

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Question # 3 of 30 (Start time: 12:11:57 PM, 22 June 2021)

Total Marks: 1

"set()" method of list class is used to:

Select the correct option



Set the value of Null Nodes



Set the value of object

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com



Set the value of lvalue



Set the value of pointer

Question # 4 of 30 (Start time: 12:12:26 PM, 22 June 2021)

Total Marks: 1

In singly linked list a node consists of two parts:

Select the correct option



Object and structure



Two pointers



Object and pointer

Solved By: Waqas Ahmed
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Two objects

Question # 5 of 30 (Start time: 12:13:07 PM, 22 June 2021)

Total Marks: 1

Suppose we are building the AVL tree using nodes 11, 29 and 35 then which type of rotation at root node will balance the tree?

Select the correct option



Single right



Double right-left



Double left-right



Single left

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Solved By: Waqas Ahmed
Whatsapp: 03066295623
Email: waqas.vu.edu@gmail.com

Question # 6 of 30 (Start time: 12:13:37 PM, 22 June 2021)

Total Marks: 1

Which kind of data structure is used to evaluate postfix expression?

Select the correct option

☒	Stack	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Queue	
☐	Linked List	
☐	Tree	

Question # 7 of 30 (Start time: 12:14:20 PM, 22 June 2021)

Total Marks: 1

In simple or singly linked list there is/are _____ pointer/s in each node.

Select the correct option



Four



Three



Two



One

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com

Question # 8 of 30 (Start time: 12:14:42 PM, 22 June 2021)

Total Marks: 1

After evaluating this expression what will be the result?
 $7+3*3/(7-4)$

Select the correct option

☐	6
☐	9
☐	12
☒	10

Solved By: Waqas Ahmed
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Email: waqas.vu.edu@gmail.com

Question # 9 of 30 (Start time: 12:15:06 PM, 22 June 2021)

Total Marks: 1

New items are added at the _____ of the stack.

Select the correct option

☐	Center	
☒	Top	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Bottom	
☐	Middle	

Question # 10 of 30 (Start time: 12:15:30 PM, 22 June 2021)

Total Marks: 1

Which data structure is needed to convert infix expression to postfix expression?

Select the correct option

☐	Tree	
☒	Stack	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Graph	
☐	Queue	

Question # 11 of 30 (Start time: 12:16:01 PM, 22 June 2021)

Total Marks: 1

A _____ model attempts to model a real-world phenomenon

Select the correct option



Conceptual



Physical



Simulation

Solved By: Waqas Ahmed

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Logical

Question # 12 of 30 (Start time: 12:16:27 PM, 22 June 2021)

Total Marks: 1

There is no such node whose next field is NULL, which one of the given option supports the statement.

Select the correct option



Array



Queue



Linked list



Circular link list

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com

Question # 13 of 30 (Start time: 12:16:50 PM, 22 June 2021)

Total Marks: 1

The abstract data type refers to the basic mathematical concept that defines the

Select the correct option



Data type

Solved By: Waqas Ahmed

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Function type



structure



Class type

Question # 14 of 30 (Start time: 12:17:15 PM, 22 June 2021)

Total Marks: 1

"new int[11]" will allocate memory for _____ integers.

Select the correct option



13



12



10



11

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com

Question # 15 of 30 (Start time: 12:17:43 PM, 22 June 2021)

Total Marks: 1

next() method of List class is used to:

Select the correct option



Moves the current position forward one element

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Email: waqas.vu.edu@gmail.com



Moves the "current" pointer to two steps after the last element of the array



Moves the "current" pointer to two steps before the last element of the array



Moves the current position backward one element

Question # 16 of 30 (Start time: 12:18:12 PM, 22 June 2021)

Total Marks: 1

Which of the following traversal method traverses the binary tree in sorted order?

Select the correct option

[Reload Math Equations](#)

Level order



Pre-order



In-order



Post-order

Solved By: Waqas Ahmed
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Question # 17 of 30 (Start time: 12:18:36 PM, 22 June 2021)

Total Marks: 1

NULL is an invalid address and _____.

Select the correct option



Points to the start point of the list



Inaccessible

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Email: waqas.vu.edu@gmail.com



Accessible



Points to the last current node of the list

Question # 18 of 30 (Start time: 12:19:07 PM, 22 June 2021)

Total Marks: 1

isFull() method of stack class will return true when:

Select the correct option



Stack is empty



Stack is full

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Stack is partially full



Stack is not empty nor full

Question # 19 of 30 (Start time: 12:19:32 PM, 22 June 2021)

Total Marks: 1

In Left-Right case of rotation in _____ tree. A double rotation is performed.

Select the correct option

☒	AVL	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Binary Search	
☐	Big	
☐	ALV	

Question # 20 of 30 (Start time: 12:20:02 PM, 22 June 2021)

Total Marks: 1

If we want to return first element of the list which one method will be used.

Select the correct option



Get();



Start();

Solved By: Waqas Ahmed

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Peek();



Set();

Question # 21 of 30 (Start time: 12:20:29 PM, 22 June 2021)

Total Marks: 1

Binary Search Tree violates the condition of AVL tree when any node has balance factor equal to

Select the correct option

Reload Math Equations



0



2 or -2

Solved By: Waqas Ahmed

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1



-1

Question # 22 of 30 (Start time: 12:20:58 PM, 22 June 2021)

Total Marks: 1

Head node pointer always points to the _____ of the linked list?

Select the correct option

[Reload Math Equations](#)

newly added node



first node

Solved By: Waqas Ahmed

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current node



last node

Question # 23 of 30 (Start time: 12:21:25 PM, 22 June 2021)

Total Marks: 1

Leaf node of binary search tree contains _____

Select the correct option

[Reload Math Equations](#)

☒	Two Null pointers	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Three Null pointers		
☐	One Null pointer		
☐	No Null pointer		

Question # 24 of 30 (Start time: 12:21:50 PM, 22 June 2021)

Total Marks: 1

Which of the following is the correct conversion of infix to postfix expression?
 $Z+B-(D-H)/K$

Select the correct option

 $ZB+DH-K-/$  $ZB+DH-K/-$

Solved By: Waqas Ahmed

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 $ZB+DHK--/$  $ZB+D-H-K/$

Question # 25 of 30 (Start time: 12:22:17 PM, 22 June 2021)

Total Marks: 1

We can use Binary Search Tree with _____

Select the correct option

☒	Both Integer and String data	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Integer data only	
☐	String data only	
☐	Non-integer data only	

Question # 26 of 30 (Start time: 12:22:46 PM, 22 June 2021)

Total Marks: 1

The -- is a decrement operator in C++ that decreases the value of the operand by _____.

Select the correct option



Three



One

Solved By: Waqas Ahmed

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Two



Four

Question # 27 of 30 (Start time: 12:23:09 PM, 22 June 2021)

Total Marks: 1

The function calls are made with the help of_____.

Select the correct option



Dynamic memory



Stack

Solved By: Waqas Ahmed

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Heap

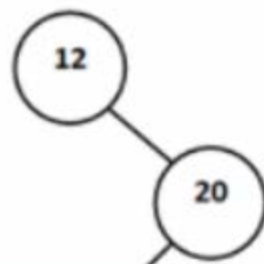


External memory

Question # 28 of 30 (Start time: 12:23:38 PM, 22 June 2021)

Total Marks: 1

Which type of rotation can balance the following AVL tree?



Select the correct option

[Reload Math Equations](#)☐

Single left

☒

Single right

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com

☐

Double left-right

☐

Double right-left

Question # 29 of 30 (Start time: 12:24:10 PM, 22 June 2021)

Total Marks: 1

In _____ various cells of memory are not located continuously.

Select the correct option



Circular array



List



Arrays



Linked list

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com

Question # 1 of 30 (Start time: 12:37:37 PM, 22 June 2021)

Linked List use _____ to store data.

Select the correct option



Variables



Array



2-D Array



Linked Memory

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com

Question # 2 of 30 (Start time: 12:38:01 PM, 22 June 2021)

In the statement template <T>, T represents_____.

Select the correct option



String data type



Character data type



Integer data type



Generic data type

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com

Question # 3 of 30 (Start time: 12:38:19 PM, 22 June 2021)

What is the postfix form of $(A+B)*(C-D)$

Select the correct option



$CD - A B + *$



$A B + C D - *$

Solved By: Waqas Ahmed

Whatsapp: 03066295623

Email: waqas.vu.edu@gmail.com



$+A B - C D *$



$A + B * C - D$



Question # 4 of 30 (Start time: 12:38:34 PM, 22 June 2021)

A queue is a _____ data structure, whereas a stack is a _____ data structure.

Select the correct option



both of these



none of these



LIFO,FIFO



FIFO, LIFO

Solved By: Waqas Ahmed

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Question # 5 of 30 (Start time: 12:38:50 PM, 22 June 2021)

The expression $DE+H^*$ is called_____

Select the correct option



Infix expression



Postfix expression

Solved By: Waqas Ahmed

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Email: waqas.vu.edu@gmail.com



Hybrid expression



Prefix expression

Question # 6 of 30 (Start time: 12:39:09 PM, 22 June 2021)

For String-based Binary Search Tree, we use ASCII values of characters for comparing among letters. This method is known as _____

Select the correct option



Lexicographic order

Solved By: Waqas Ahmed
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Heap-based approach



Alphabet coding procedure



Asymmetric technique

Question # 7 of 30 (Start time: 12:39:27 PM, 22 June 2021)

Current pointer always points to the _____ of the linked list.

Select the correct option



lastcurrent node



last node



first node



current node

Solved By: Waqas Ahmed

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Question # 8 of 30 (Start time: 12:39:40 PM, 22 June 2021)

Which one option is correct to know the first element of the linked list.

Select the correct option

☒	Start(); Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Current pointer
☐	Last pointer
☐	Head pointer

Question # 9 of 30 (Start time: 12:39:56 PM, 22 June 2021)

Suppose we are building the AVL tree using nodes 11, 29 and 35 then which type of rotation at root node will balance the tree?

Select the correct option



Single left

Solved By: Waqas Ahmed
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Double left-right



Single right



Double right-left

Question # 10 of 30 (Start time: 12:40:18 PM, 22 June 2021)

The type of expression in which operator succeeds its operands is _____ expression.

Select the correct option

☐	Preorder
☐	Prefix
☐	Infix
☒	Postfix

Solved By: Waqas Ahmed
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Email: waqas.vu.edu@gmail.com

Question # 11 of 30 (Start time: 12:40:32 PM, 22 June 2021)

All the objects created using _____ operator have to be destroyed explicitly.

Select the correct option



construct



new

Solved By: Waqas Ahmed

Whatsapp: 03066295623

Email: waqas.vu.edu@gmail.com



build



create

Question # 12 of 30 (Start time: 12:40:46 PM, 22 June 2021)

A binary tree is said to be a _____ binary tree if every non-leaf node in a binary tree has non-empty left and right subtrees.

Select the correct option



Strictly

Solved By: Waqas Ahmed

Whatsapp: 03066295623

Email: waqas.vu.edu@gmail.com



Complete



AVL



Perfect

Question # 13 of 30 (Start time: 12:41:03 PM, 22 June 2021)

Memory address is stored in

Select the correct option



Reference



Pointer

Solved By: Waqas Ahmed

Whatsapp: 03066295623

Email: waqas.vu.edu@gmail.com



Array



Address operator

Question # 14 of 30 (Start time: 12:41:22 PM, 22 June 2021)

In which of the following tree for each node, the value of root node is greater than left child and smaller than right child ?



Select the correct option



Binary Search Tree

Solved By: Waqas Ahmed

Whatsapp: 03066295623

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Full Binary Tree



Perfect Binary Tree



Complete Binary Tree

Question # 15 of 30 (**Start time: 12:41:39 PM, 22 June 2021**)

The _____ method of list will position the currentNode and lastCurrentNode at the start of the list.

Select the correct option

☒	Start Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Next
☐	Remove
☐	Back

Question # 16 of 30 (Start time: 12:41:57 PM, 22 June 2021)

An array is collection of cells of the .

Select the correct option



Same type

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Different type



Hybrid type



Struct type

Question # 17 of 30 (Start time: 12:42:14 PM, 22 June 2021)

In a complete binary tree the number of nodes at level 5 are

Select the correct option



16



32

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31



15

Question # 18 of 30 (Start time: 12:42:29 PM, 22 June 2021)

The _____ of a binary tree is the maximum level of the deepest leaf in the tree.

Select the correct option



Height



Length



Width



Depth

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Question # 19 of 30 (Start time: 12:42:42 PM, 22 June 2021)

Circular Linked List always has _____ NULL pointer/s in a node.

Select the correct option



1



0

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2



3

Question # 20 of 30 (Start time: 12:42:59 PM, 22 June 2021)

In the perspective of memory organization every process executing, the last part of the memory is for the _____ of the program.

Select the correct option



Data



Stack



Code



Heap

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Question # 21 of 30 (Start time: 12:43:13 PM, 22 June 2021)

Which of the following is correct syntax to declare a const function ?

Select the correct option

☒ `int& findMin( ) const;`

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☐ `const int& findMin( );`

☐ `int& findMin(const);`

☐ `int& findMin const ( );`

Question # 22 of 30 (Start time: 12:43:31 PM, 22 June 2021)

The abstract data type refers to the basic mathematical concept that defines the

Select the correct option



Function type



Class type



Data type

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Email: waqas.vu.edu@gmail.com



structure

Question # 23 of 30 (Start time: 12:43:47 PM, 22 June 2021)

_____ is a self-balancing tree.

Select the correct option

☐	ALV	
☒	AVL	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Binary Tree	
☐	Binary Search Tree	

Question # 24 of 30 (Start time: 12:44:04 PM, 22 June 2021)

Whenever we call a function, the compiler makes a stack, the top element of the stack is _____ of the function.

Select the correct option

☒	Return address	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	First argument	
☐	Last argument	
☐	Variable name only	

Question # 25 of 30 (Start time: 12:44:25 PM, 22 June 2021)

Which of the following is TRUE about recursive calls ?

Select the correct option



Recursive calls can only be used with inorder traversal



Recursive call always contains a terminating condition

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Tree data structure cannot use recursive calls



There is no terminating condition in recursive calls

Question # 26 of 30 (Start time: 12:44:45 PM, 22 June 2021)

What will be the postfix expression of following infix expression?
 $D+E*F/G$

Select the correct option



$DE+F*G/$



$DEF*G/+$

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Email: waqas.vu.edu@gmail.com



$DE*F/G+$



$DE+FG*/$

Question # 27 of 30 (Start time: 12:45:02 PM, 22 June 2021)

_____ is a collection of elements arranged in a linear order.

Select the correct option



Stack

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Binary Search Tree



AVL Tree



Heap

Question # 28 of 30 (Start time: 12:45:18 PM, 22 June 2021)

In a list, tail() method of current pointer _____

Select the correct option



returns the first element of the "current" pointer



moves the "current" pointer to the very first element



moves the "current" pointer to the very last element

Solved By: Waqas Ahmed

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returns the last element of the "current" pointer

Question # 29 of 30 (Start time: 12:45:32 PM, 22 June 2021)

After performing which task on the list, implementing array has to shift the entire remaining elements one place to the left.

Select the correct option

☒	By removing element at start	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	By adding element at last	
☐	By searching of element	
☐	By removing element at end	

Question # 30 of 30 (Start time: 12:45:46 PM, 22 June 2021)

If a function has recursive call as the last statement, it is known as _____.

Select the correct option



Last recursion



Local Recursion



Tail recursion

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Function recursion

Question # 1 of 30 (Start time: 09:26:13 AM, 22 June 2021)

Total Marks: 1

_____ is a collection of elements arranged in a linear order.

Select the correct option



AVL Tree



Binary Search Tree



Stack

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Email: waqas.vu.edu@gmail.com



Heap

Question # 2 of 30 (Start time: 09:26:40 AM, 22 June 2021)

Total Marks: 1

Analyze the given code carefully and identify which type of binary tree traversal this is :

```
void traversal (TreeNode<int>* treeNode)
{
    if( treeNode != NULL )
    {
        traversal (treeNode->getLeft());
    }
}
```

Select the correct option

[Reload Math Equations](#)

Levelorder



Preorder



Inorder

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Postorder

Question # 3 of 30 (Start time: 09:27:13 AM, 22 June 2021)

Total Marks: 1

In the linked list implementation of the stack class, where does the push member function places the new entry on the linked list?

Select the correct option



After all other entries that are greater than the new entry.



At the head

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After all other entries that are smaller than the new entry.



At the tail

Question # 4 of 30 (Start time: 09:27:44 AM, 22 June 2021)

Total Marks: 1

Sometimes, _____ queues do not follow FIFO order

Select the correct option

[Reload Math Equations](#)

☒	Priority	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Simple	
☐	Circular	
☐	Double Ended	

Question # 5 of 30 (Start time: 09:28:15 AM, 22 June 2021)

Total Marks: 1

Clear() of a list is used to_____.

Select the correct option

Reload Math Equations



remove last element



remove current element



remove all elements

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Email: waqas.vu.edu@gmail.com



remove first element

Question # 6 of 30 (Start time: 09:28:44 AM, 22 June 2021)

Total Marks: 1

Which data structure is needed to convert infix expression to postfix expression?

Select the correct option



Stack

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Email: waqas.vu.edu@gmail.com



Graph



Queue



Tree

Question # 7 of 30 (Start time: 09:29:11 AM, 22 June 2021)

Total Marks: 1

The usage of the class of a data structure causes the code of the program

Select the correct option

☒	Simplifies	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Difficult	
☐	Complicated	
☐	Lengthy	

Question # 8 of 30 (Start time: 09:29:38 AM, 22 June 2021)

Total Marks: 1

If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?

Select the correct option



222



48



5

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Email: waqas.vu.edu@gmail.com



4

Question # 9 of 30 (Start time: 09:30:09 AM, 22 June 2021)

Total Marks: 1

The _____ of a node in a binary tree is defined as the height of its left subtree minus height of its right subtree.

Select the correct option



Width



Level



Height



Balance

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Email: waqas.vu.edu@gmail.com

Question # 10 of 30 (Start time: 09:30:35 AM, 22 June 2021)

Total Marks: 1

copy() method of List data structure _____

Select the correct option

☒	Set one list to be a copy of another	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Copy any item of list	
☐	Copy last item of list	
☐	Copy first item of list	

Question # 11 of 30 (Start time: 09:31:16 AM, 22 June 2021)

Total Marks: 1

In Binary Search Tree, deleting a node is easy if it is a _____ node.

Select the correct option



Left-most



Root



Leaf

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Right-most

Question # 12 of 30 (Start time: 09:31:45 AM, 22 June 2021)

Total Marks: 1

STL is a _____ that is a part of the official standard of C++.

Select the correct option



C program file



.h file



Library

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.cpp file

Question # 13 of 30 (Start time: 09:32:15 AM, 22 June 2021)

Total Marks: 1

Which of the following is the correct conversion from infix to postfix expression?
 $A*B+C/(E-F)$

Select the correct option

 $AB+C*E-F/$  $AB*CEF-/+$

Solved By: Waqas Ahmed

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 $AB*C+EF/-$  $ABC*+EF-/$

Question # 14 of 30 (Start time: 09:32:49 AM, 22 June 2021)

Total Marks: 1

Which kind of data structure is used to evaluate postfix expression?

Select the correct option

☒	Stack	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Linked List	
☐	Tree	
☐	Queue	

Question # 15 of 30 (Start time: 09:33:31 AM, 22 June 2021)

Total Marks: 1

In singly linked list "next" field of node contains:

Select the correct option

☒	Address of next node	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Address of head node	
☐	Object of next node	
☐	Object of current node	

Question # 16 of 30 (Start time: 09:34:02 AM, 22 June 2021)

Total Marks: 1

The balance of a node in a binary tree is defined as the height of its _____ sub tree minus height of its right sub tree.

Select the correct option

☐	upper	
☒	left	Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	lower	
☐	right	

Question # 17 of 30 (Start time: 09:34:34 AM, 22 June 2021)

Total Marks: 1

```
y = &x[0];
```

In the above statement, we get the address of the first location of the array x and store it in y. Here "y" is:

Select the correct option



lvalue

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zvalue



xvalue



rvalue

Question # 18 of 30 (Start time: 09:35:05 AM, 22 June 2021)

Total Marks: 1

In a list, tail() method of current pointer _____

Select the correct option



returns the first element of the "current" pointer



moves the "current" pointer to the very last element

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returns the last element of the "current" pointer



moves the "current" pointer to the very first element

Question # 19 of 30 (Start time: 09:35:32 AM, 22 June 2021)

Total Marks: 1

get(?) method of List class is used to:

Select the correct option



Get element from the last position



Get element from the middle position



Get element at the given position



Get element from the first position

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Question # 20 of 30 (Start time: 09:35:57 AM, 22 June 2021)

Total Marks: 1

If the root of a tree is at level zero, its two children (subtrees) i.e. nodes will be at _____

Select the correct option

☐	Level 3
☐	Level 0
☐	Level 2
☒	Level 1

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Question # 21 of 30 (Start time: 09:36:22 AM, 22 June 2021)

Total Marks: 1

Which of the following statement is correct for the variable "current--"?

Select the correct option

 $\text{current} = \text{current} - 1$

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Email: waqas.vu.edu@gmail.com

 $\text{current} = \text{current} - 2$  $\text{current} = \text{current} + 1$  $\text{current} - 1 = \text{current}$

Question # 22 of 30 (Start time: 09:37:11 AM, 22 June 2021)

Total Marks: 1

Suppose we are building the AVL tree using nodes 1, 23 and 12 then which type of rotation at root node will balance the tree?

Select the correct option



Single left



Double right-left

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Single right



Double left-right

Question # 23 of 30 (Start time: 09:37:50 AM, 22 June 2021)

Total Marks: 1

Last node in circular linked list contains

Select the correct option



At least one null pointer



tail pointer



Maximum two null pointers



No null pointer

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Question # 24 of 30 (Start time: 09:38:15 AM, 22 June 2021)

Total Marks: 1

If we return the reference of a local variable from a function it will cause :

Select the correct option



dangling reference

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deletion of local variable from memory



reference overloading



duplication of local variable

Question # 25 of 30 (Start time: 09:38:59 AM, 22 June 2021)

Total Marks:

_____ is the major factor to see the efficiency of a program.

Select the correct option

☐	Correctness
☐	Quality
☐	Code
☒	Time

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Question # 26 of 30 (Start time: 09:39:26 AM, 22 June 2021)

Total Marks: 1

To create a reference variable, we need to use _____ sign.

Select the correct option

[Reload Math Equations](#)

asterisk



sigma



dollar



ampersand

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Question # 27 of 30 (Start time: 09:39:54 AM, 22 June 2021)

Total Marks: 1

The _____ tree has been named after two persons.

Select the correct option

[Reload Math Equations](#)

Red Black



Binary Search



AVL

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Email: waqas.vu.edu@gmail.com



Binary

Question # 28 of 30 (Start time: 09:40:20 AM, 22 June 2021)

Total Marks: 1

Following is true in case of using Recursive method calls.

Select the correct option

[Reload Math Equations](#)

Code will not execute if it contains recursive calls



The code becomes very short

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There is no effect on length of code



The code becomes very long

Question # 30 of 30 (Start time: 09:40:50 AM, 22 June 2021)

Total Marks: 1

Suppose we have the following values to be inserted in constructing AVL tree,
20,23,25,10,12,13
Tell when first rotation will take place,

Select the correct option



after inserting node 10



after inserting node 25



after inserting node 12

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after inserting node 23

CS301:Grand Quiz

Question # 1 of 30 (Start time: 12:09:40 PM, 22 June 2021)

dequeue() operation of queue data structure is used to _____

Select the correct option



insert an element at the front



remove an element from the front and return it

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Email: waqas.vu.edu@gmail.com



insert an element at the back



get an element from the front of the queue

Question # 2 of 30 (Start time: 12:10:58 PM, 22 June 2021)

In queue data structure elements are removed from _____.

Select the correct option

☐	Pop
☐	Rear
☐	Push
☒	Front

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Email: waqas.vu.edu@gmail.com

Question # 3 of 30 (Start time: 12:11:49 PM, 22 June 2021)

Suppose we are building the AVL tree using nodes 13, 7 and 9 then which type of rotation at root node will balance the tree?

Select the correct option

☐	Single left
☒	Double left-right Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	Double right-left
☐	Single right

Question # 4 of 30 (Start time: 12:12:10 PM, 22 June 2021)

Which of the following operation returns but do not removes top value of the stack?

Select the correct option

☐	push
☒	top Solved By: Waqas Ahmed Whatsapp: 03066295623 Email: waqas.vu.edu@gmail.com
☐	first
☐	pop

Question # 6 of 30 (Start time: 12:13:54 PM, 22 June 2021)

Which of the following is not a form of expression?

Select the correct option



Postfix



Infix



Prefix



Pastfix

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Question # 7 of 30 (Start time: 12:14:20 PM, 22 June 2021)

get(?) method of List class is used to:

Select the correct option



Get element from the first position



Get element from the last position



Get element from the middle position



Get element at the given position

Solved By: Waqas Ahmed

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Question # 8 of 30 (Start time: 12:14:50 PM, 22 June 2021)

Consider the linked list implementation of a stack. Which of the following node is considered as Top of the stack?

Select the correct option



Middle node



Last node



Any node



First node

Solved By: Waqas Ahmed

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